

ADC to S3: prove DMA capture

ADS127L18EVM | Step-by-step plan | October 6, 2026

Goal: continuously capture at least 24 Mbit/s of useful sample data into S3 RAM. Confirm compatibility before connecting digital signals.

1. Settle the connection details

Verify the ADC clock, data format, S3 input capability, exact pins, and suitable 1.8 V / 3.3 V level shifters. Produce a wire-by-wire diagram.

Pass when: A wiring plan with no unresolved electrical or timing assumptions.

2. Power the ADC board

Connect its supply and measure its power and logic voltages before connecting signal wires.

Pass when: Correct, stable voltages.

3. Connect Feather configuration wires

Connect SDI, SDO, SCLK, CS and common ground through the required level shifting. After Karl authorizes coding, add configuration code.

Pass when: Feather reads the ADC identity and verifies settings by reading them back.

4. Configure one channel

Enable Channel 0, disable the others, select the low-latency filter, and initially use a slower sample rate.

Pass when: The ADC generates the expected data clock and sample-boundary signals. Verify with a scope or suitable logic analyzer when available.

ADC to S3: prove DMA capture

ADS127L18EVM | Step-by-step plan | October 6, 2026

Test scope: capture, check and reuse RAM buffers. Add SD recording after this capture path passes.

5. Connect ADC output to S3

Connect DCLK, FSYNC, DOUT0 and ground through suitable level shifting. Implement capture into RAM using DMA (Direct Memory Access).

Pass when: Hardware fills memory buffers. The processor handles completed buffers rather than every individual sample.

6. Verify the captured numbers

Use an ADC test pattern if available, then a known analog signal. Check sample boundaries, bit order and positive/negative readings.

Pass when: Correct data. An increasing sample counter alone does not establish correctness.

7. Increase to the target rate

Raise acquisition to approximately one million samples/second. Process and reuse completed RAM buffers continuously.

Pass when: Sustained target throughput without overwritten buffers or reported capture errors.

8. Run a sustained test

Start with one minute, then 30 minutes. Report samples/second, completed buffers, overruns and data-check failures.

Pass when: Stable throughput and zero detected errors throughout the run. A known pattern provides stronger evidence of missing or corrupted data than an analog waveform.

Rate accounting: $1,024,000 \text{ samples/s} \times 24 \text{ useful bits} = 24.576 \text{ Mbit/s}$ (3.072 MB/s). Clock overhead and RAM word padding may require additional bandwidth; step 1 must establish the actual requirements.

Reference: TI ADS127L1x datasheet. The proposed ADC-to-S3 interface remains unverified.